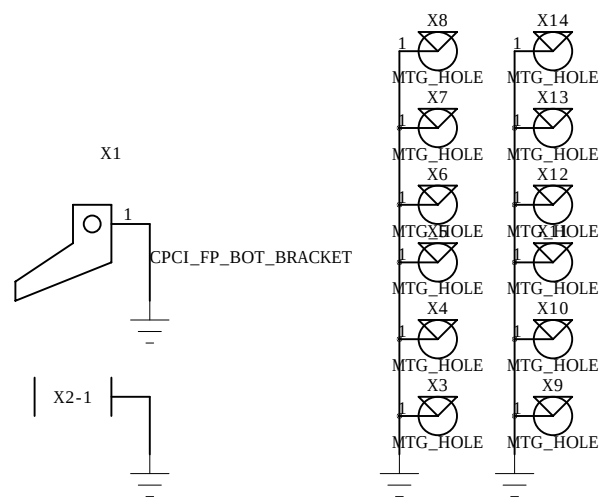
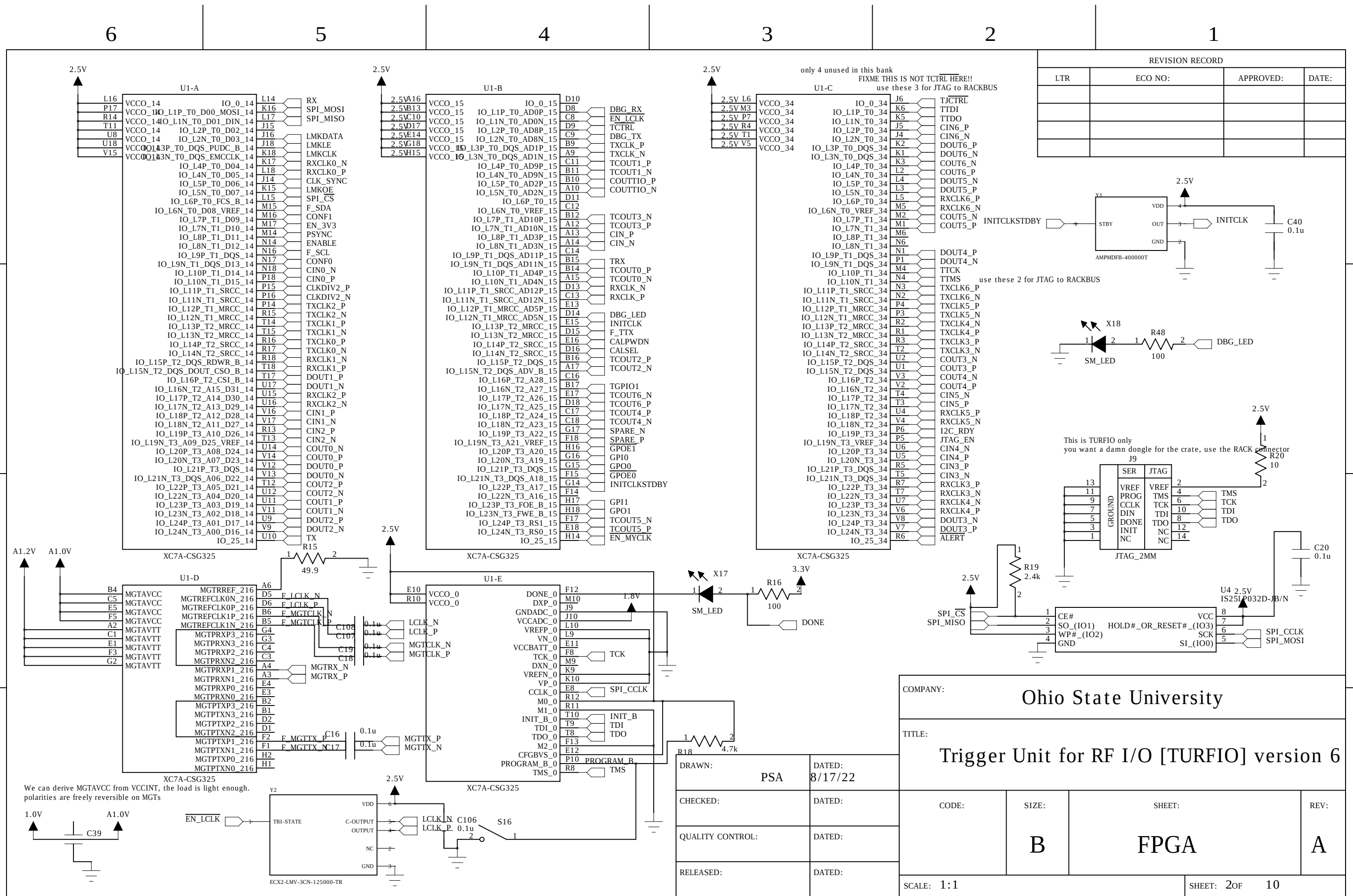


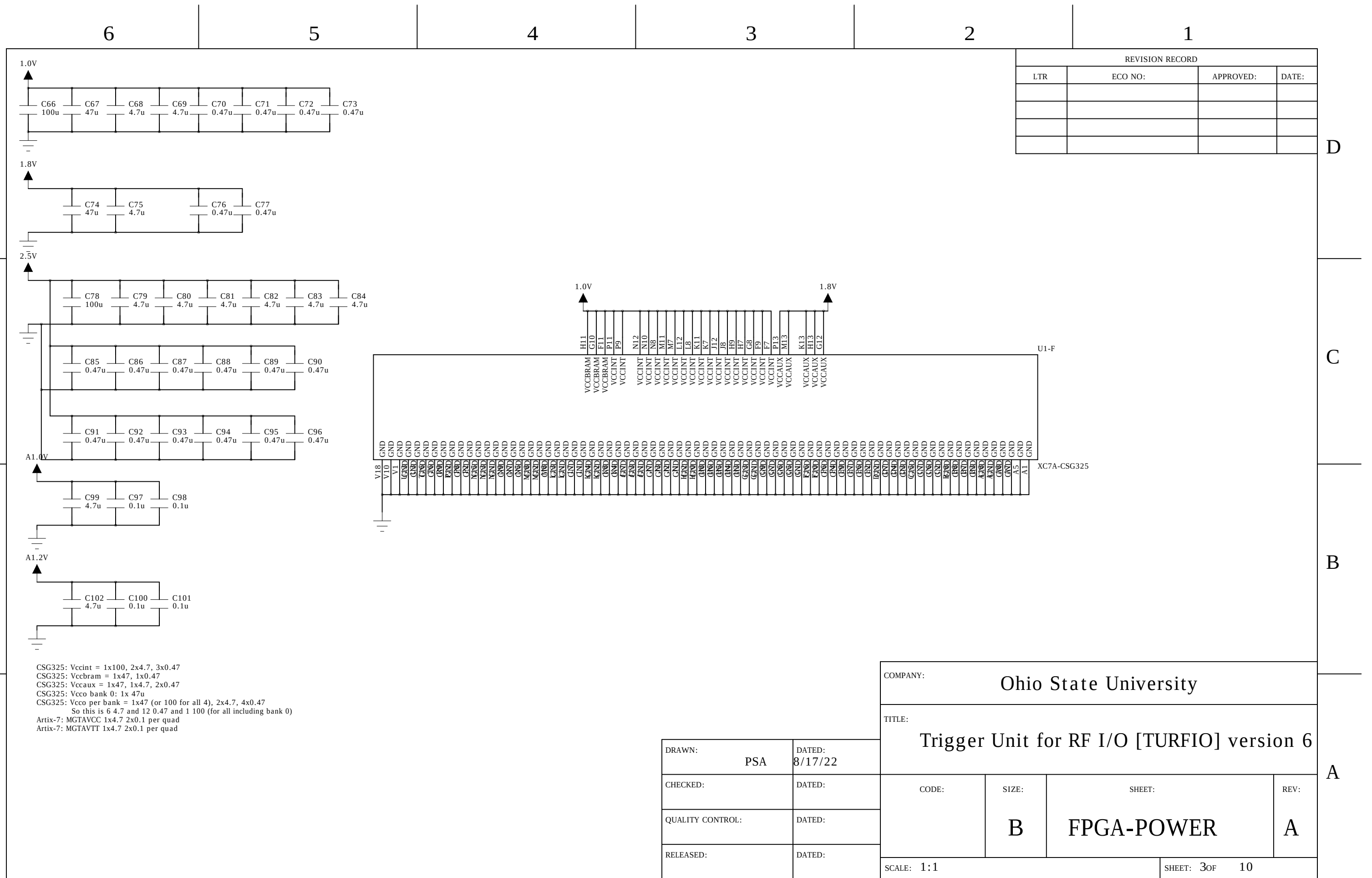


REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

Allpcb 6 layer stackup has 7.28 mil top/layer 2 spacing
7/4 is close enough (106-110 ohms)
For the GTP links we do 8/4 (or 7.67/4) which is closer
Microstrip is 12 mil

DRAWN: PSA		DATED: 8/17/22		Trigger Unit for RF I/O [TURFIO] version 6									
CHECKED:		DATED:		CODE:		SIZE: B		SHEET: TOP				REV: A	
QUALITY CONTROL:		DATED:											
RELEASED:		DATED:		SCALE: 1:1					SHEET: 1 OF 10				





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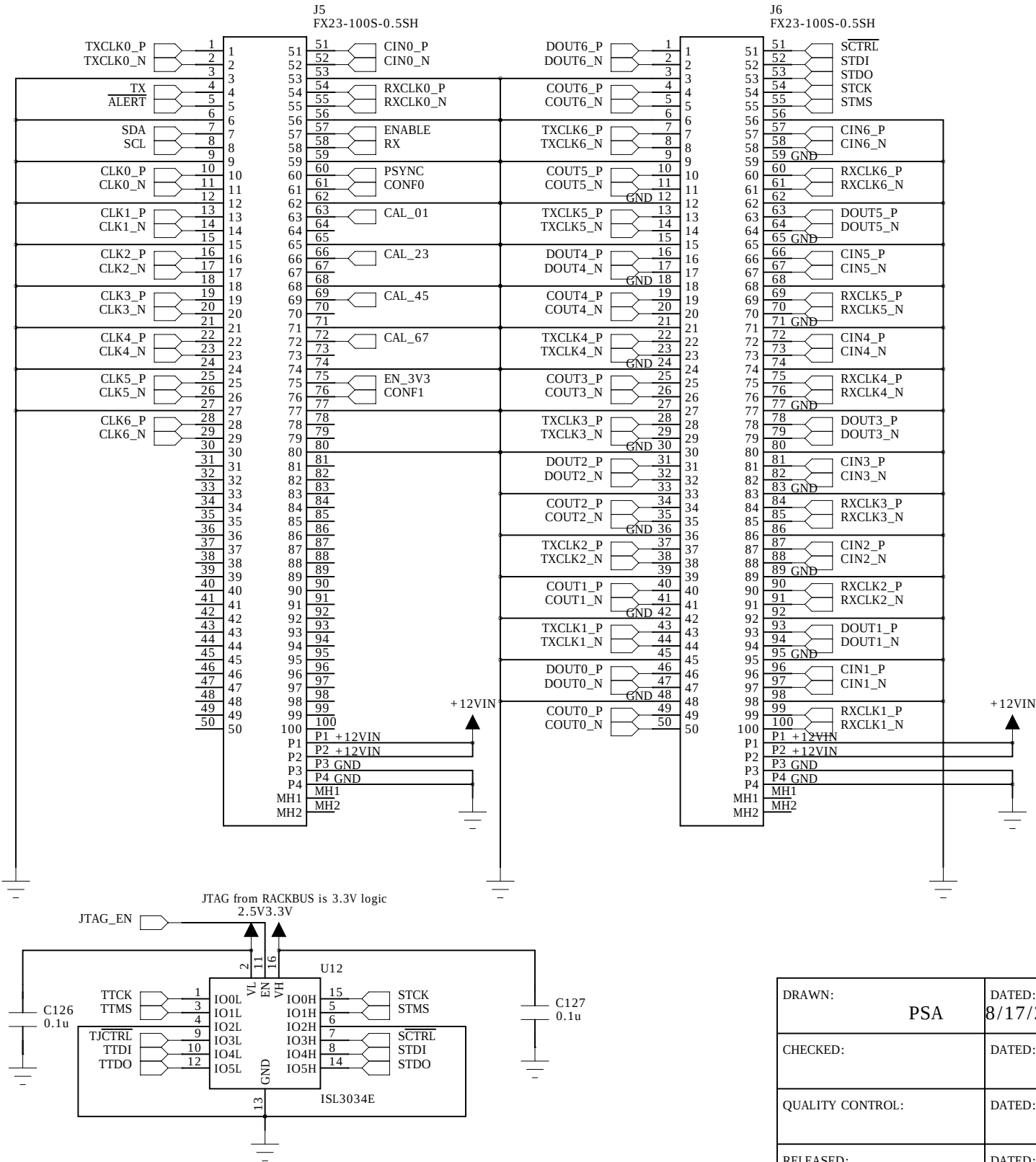
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REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

Clocks are all LMK outputs
Cals are distribution outputs

CIN/COUT: 28
RXCLK/TXCLK: 28
DOUT: 14
TURF: 24
This is 94 already

We have 150, so should be fine

RX/TX/SDA/SCL/ALERT/ENABLE: 6 (2.5V okay except add series resistor to RX)
EN_3V3/CONF01/PSYNC: 4 (2.5V okay)
JTAG: 5 (this is 3.3V! Need level translator!)
LMK: 5 (2.5V okay)
CAL_SEL: 1-3? (2.5V okay)
TOUT/TIN: 2 (schmitt receiver/line driver)
CLK: 2

ENABLE is either sink only (if it's really ENABLE)
or the diode is bypassed and it's directly driven

Names here are all from *SURF* perspective
So TXCLKs need to go to MRCC/SRCCs

Match CIN/RXCLK and COUT/TXCLK to 1 mm (1/3 ps)

CIN6/RXCLK6: 103.65 +/- 0.05	COUT6/TXCLK6: 85.7 +/- 0.05
CIN5/RXCLK5: 81.65 +/- 0.05	COUT5/TXCLK5: 79.8 +/- 0.05
CIN4/RXCLK4: 76.2 +/- 0.05	COUT4/TXCLK4: 73.55 +/- 0.05
CIN3/RXCLK3: 75.4 +/- 0.05	COUT3/TXCLK3: 73.65 +/- 0.05
CIN2/RXCLK2: 76.85 +/- 0.05	COUT2/TXCLK2: 75.2 +/- 0.05
CIN1/RXCLK1: 80 +/- 0.1	COUT1/TXCLK1: 68.7 +/- 0.1
CIN0/RXCLK0: 86.2 +/- 0.05	COUT0/TXCLK0: 70.15 +/- 0.05

(add 3.2 mm to these for absolute) (these are all absolute)

COMPANY:			
Ohio State University			
TITLE:			
Trigger Unit for RF I/O [TURFIO] version 6			
DRAWN:	DATED:	REV:	
PSA	8/17/22	A	
CHECKED:	DATED:	REV:	
QUALITY CONTROL:	DATED:	REV:	
RELEASED:	DATED:	REV:	
SCALE: 1:1		SHEET: 4 of 10	

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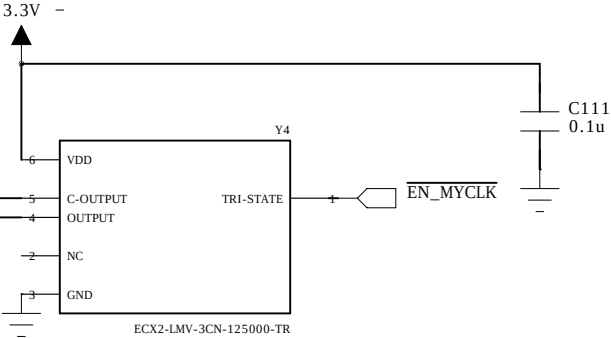
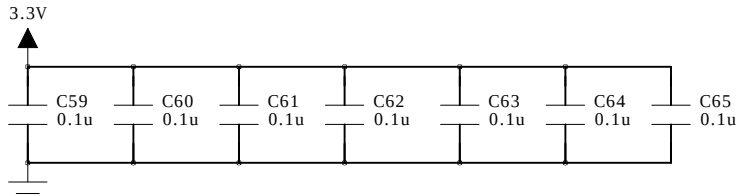
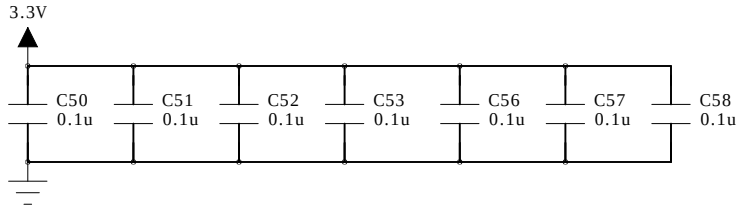
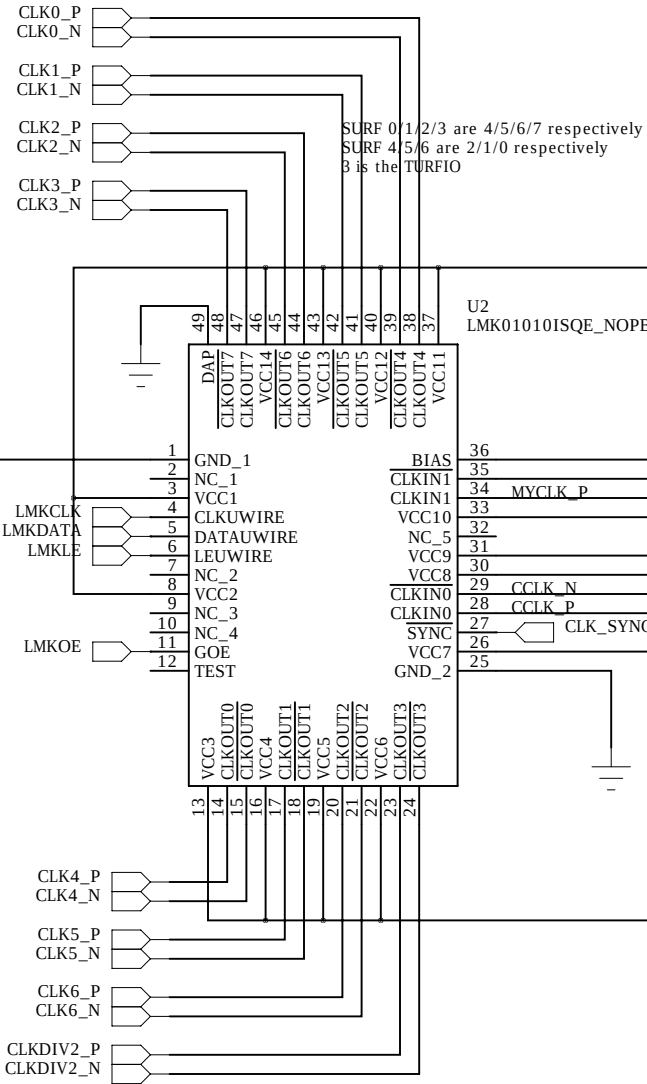
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Use LMK01010 to buffer/divide input clock
Can broadcast 125 MHz:
divide by 16 to get 7.8125 MHz, divide by 2 (min) to get 62.5 MHz for FPGA
Boost back to 125 MHz in FPGA, use commanding to synchronize dividers



COMPANY:		Ohio State University	
TITLE:		Trigger Unit for RF I/O [TURFIO] version 6	
CODE:	SIZE:	SHEET:	REV:
	B	CLOCK	A
SCALE: 1:1		SHEET: 5 of 10	

DRAWN:	PSA	DATED:	8/17/22
CHECKED:		DATED:	
QUALITY CONTROL:		DATED:	
RELEASED:		DATED:	

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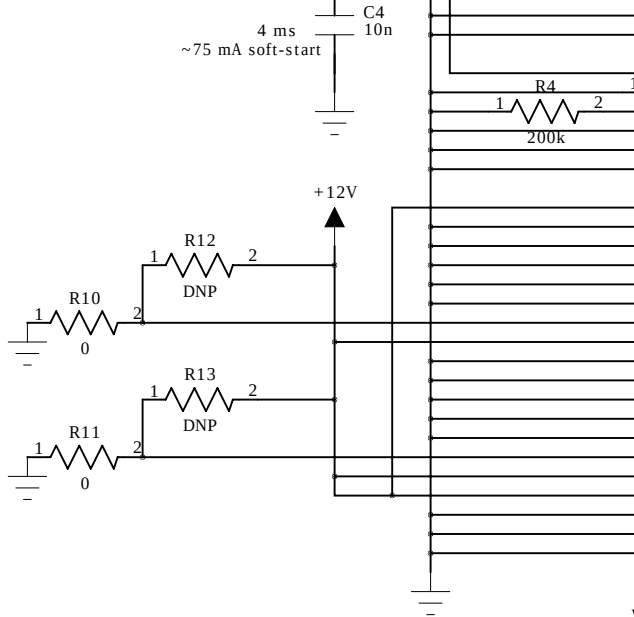
These are all overmargined by x2
We need:
1.0V (VCCINT/VCCBRAM/MGTAVCC), 0.7 A
1.8V (VCCAUX), 0.5 A
2.5V (bank I/O), 0.6A
3.3V (clock), 0.4 A
plus
1.2V (LDO from VCCAUX), 100 mA

Efficiencies are all 80%, so figure 1W here
Sequence is VCCINT, VCCAUX, VCCO.

3.3V and VCCINT turn on first (using RUN14)
VCCINT turns on the 23 converter using PG4->RUN23
Then VCCAUX turns on VCCO using PG2->TRSS3
Finally VCCO releases PROG_B to allow turnon
The simulation has screwed up PG leakage currents so these values look different.

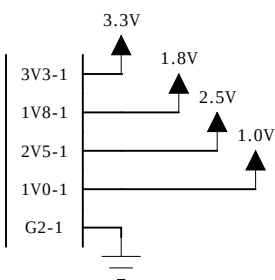
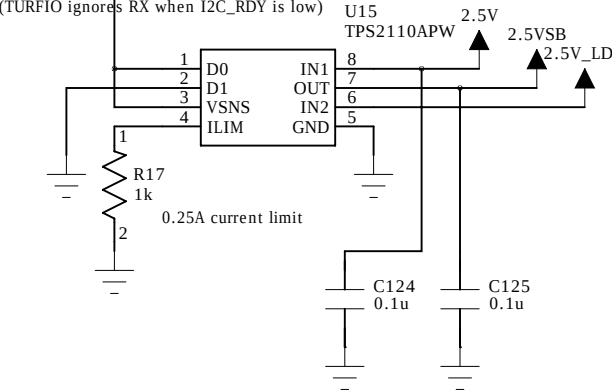
Required bulk capacitance for Artix-7:
VCCINT: 147 uF
VCCAUX: 47 uF
VCCO: 147 uF

VOUT1: 3.3V
VOUT4: 1.0V
VOUT2: 1.8V
VOUT3: 2.5V

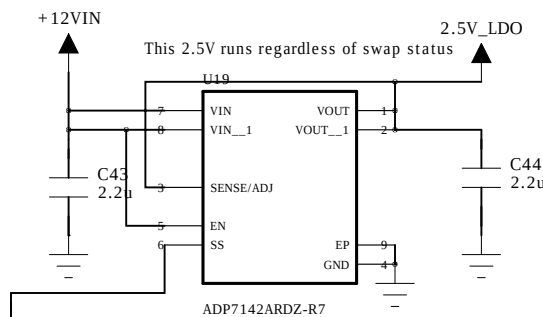
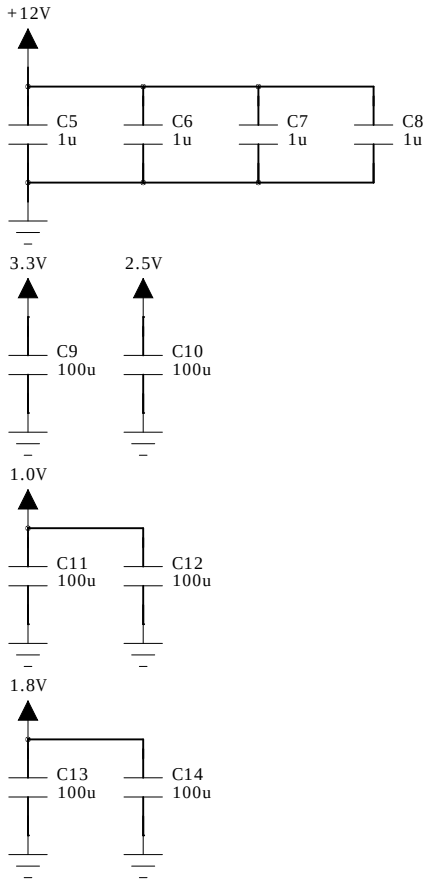


We need a 2.5VSB for TURF monitoring, so we failover generate one here
Logic we want is to choose 2.5V_LDO if PG_2V5 is low, otherwise choose 2.5V.
Use TPS2110A, tie VSNS/D0 to PG_2V5 (INIT_B) and D1 to GND.
Then tie 2.5V to IN1 and 2.5V_LDO to IN2.

We then use a pair of TCA9803s to isolate the crate I2C bus:
if TCTRL is high, TURF RX/TX controls the I2C bus, otherwise TURFIO does
Need a 1-bit buffer to disconnect TX as well
(TURFIO ignores RX when I2C_RDY is low)



REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



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DRAWN:	PSA	DATED:	8/17/22
CHECKED:		DATED:	
QUALITY CONTROL:		DATED:	
RELEASED:		DATED:	

COMPANY: Ohio State University			
TITLE: Trigger Unit for RF I/O [TURFIO] version 6			
CODE:	SIZE: B	SHEET: POWER	REV: A
SCALE: 1:1			SHEET: 6 of 10

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REVISION RECORD

LTR	ECO NO:	APPROVED:	DATE:

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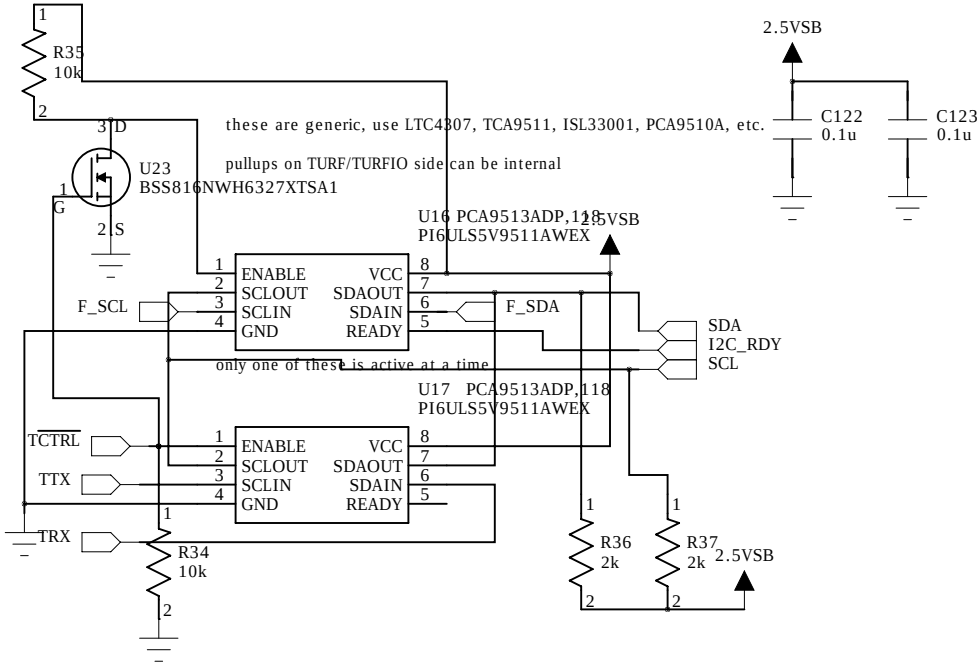
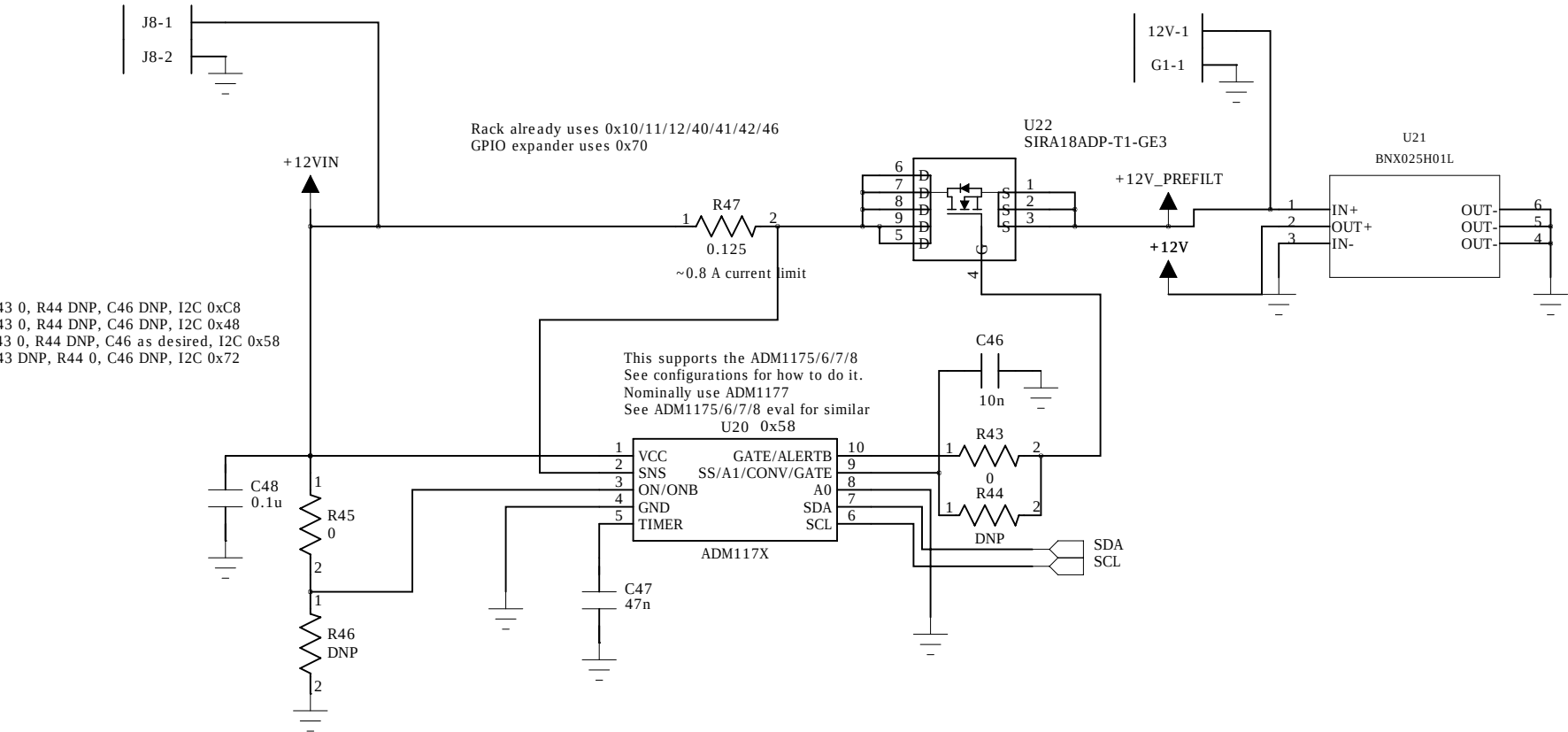
A

A

ADM1175: R43 0, R44 DNP, C46 DNP, I2C 0xC8
ADM1176: R43 0, R44 DNP, C46 DNP, I2C 0x48
ADM1177: R43 0, R44 DNP, C46 as desired, I2C 0x58
ADM1178: R43 DNP, R44 0, C46 DNP, I2C 0x72

Rack already uses 0x10/11/12/40/41/42/46
GPIO expander uses 0x70

This supports the ADM1175/6/7/8
See configurations for how to do it.
Nominally use ADM1177
See ADM1175/6/7/8 eval for similar
U20 0x58



COMPANY: Ohio State University			
TITLE: Trigger Unit for RF I/O [TURFIO] version 6			
CODE:	SIZE: B	SHEET: POWER-SWAP	REV: A
SCALE: 1:1		SHEET: 7 of 10	

DRAWN: PSA	DATED: 8/17/22
CHECKED:	DATED:
QUALITY CONTROL:	DATED:
RELEASED:	DATED:

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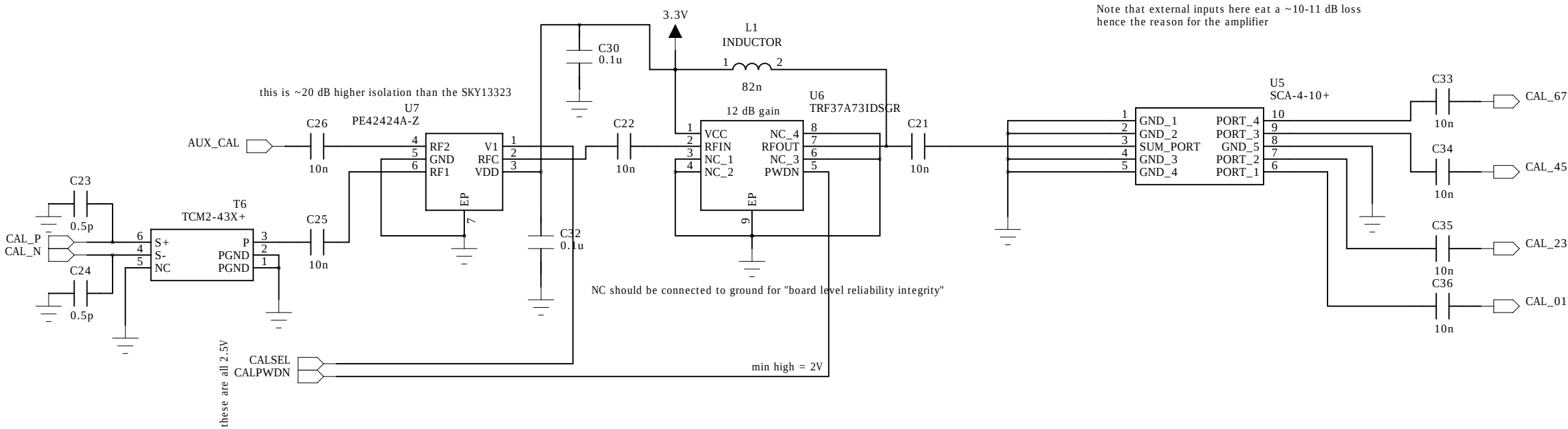
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

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COMPANY:

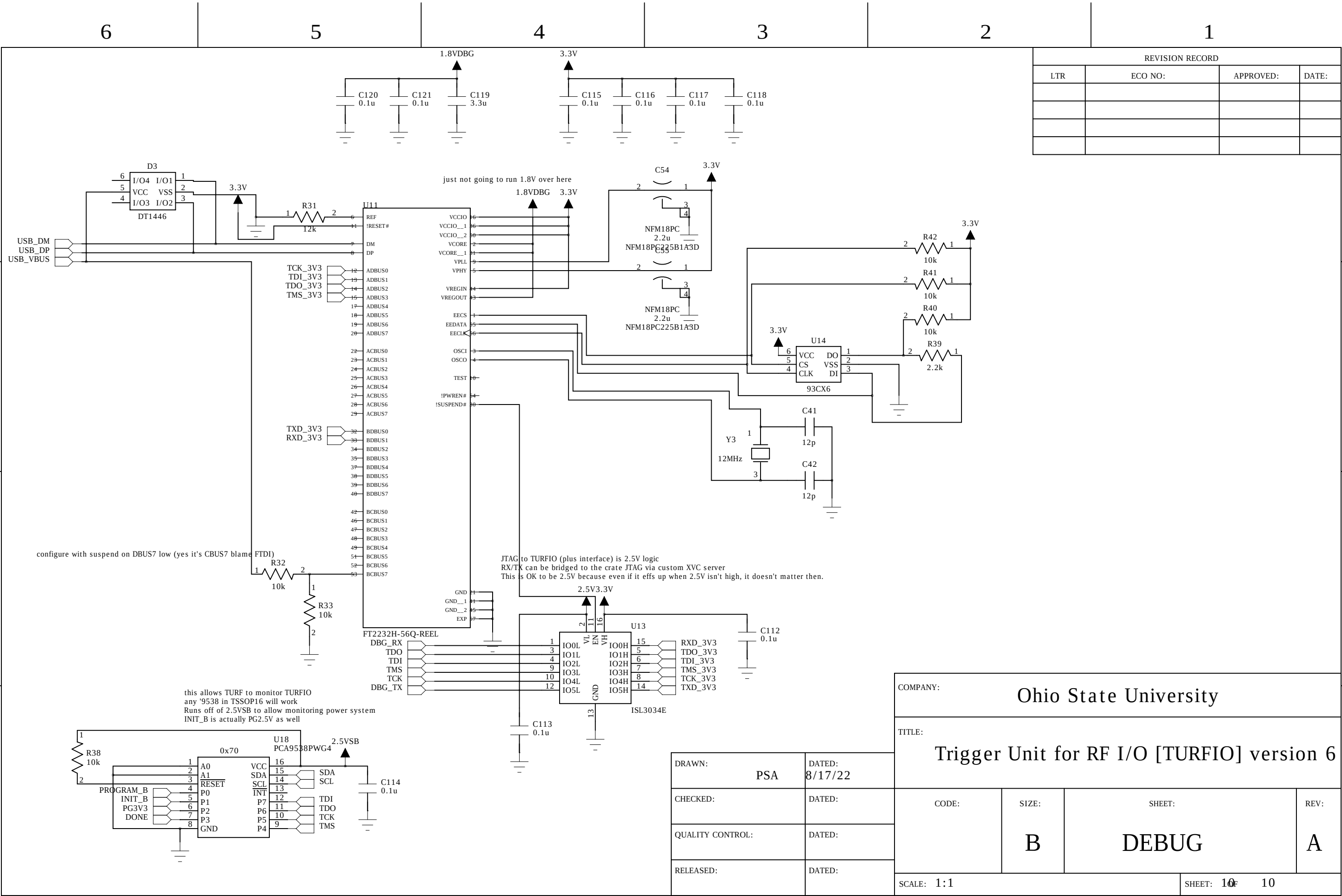
Ohio State University

TITLE:

Trigger Unit for RF I/O [TURFIO] version 6

CODE:	SIZE:	SHEET:	REV:
	B	CAL	A
SCALE: 1:1		SHEET: 9 of 10	

DRAWN:	PSA	DATED:	8/17/22
CHECKED:		DATED:	
QUALITY CONTROL:		DATED:	
RELEASED:		DATED:	



REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

COMPANY: Ohio State University			
TITLE: Trigger Unit for RF I/O [TURFIO] version 6			
CODE:	SIZE: B	SHEET: DEBUG	REV: A
SCALE: 1:1			SHEET: 10 of 10